



▼ **Figure 47.20 Determination of germ cell fate in *C. elegans*.** Labeling with a fluorescent antibody that is specific for a *C. elegans* P granule protein (green) reveals the incorporation of P granules into four cells of the newly hatched larva (two of the four cells are visible in this view).



from the fertilized egg in virtually the same way for every individual. Careful microscopic observations of worms at all stages of development, coupled with experiments in which particular cells or groups of cells were destroyed by a laser beam or through mutations, resulted in the cell lineage diagram shown in Figure 47.19. Using this cell lineage diagram, you can identify all of the progeny of a single cell, just as you would use a family history to trace the descendants of, for example, one great-great-grandparent.

As an example of a particular cell fate, let's consider *germ cells*, the specialized cells that give rise to eggs or sperm. In all animals studied, complexes of RNA and protein direct particular cells to become germ cells. In *C. elegans*, such complexes, called *P granules*, can be detected in four cells of the newly hatched larva (Figure 47.20) and, later, in the cells of the adult gonad that produce sperm or eggs.

Tracing the position of the P granules provides a dramatic illustration of how cells acquire a specific fate, during development. As shown in Figure 47.21, the P granules are distributed throughout the newly fertilized egg but move to the posterior end of the zygote before the first cleavage division. As a result, only the posterior of the two cells formed by the first division contains P granules. The P granules continue to be asymmetrically partitioned during subsequent divisions. Thus, the P granules act as cytoplasmic determinants (see Concept 18.4), fixing germ cell fate at the earliest stage of *C. elegans* development.

Fate mapping in *C. elegans* paved the way for major discoveries about programmed cell death. Lineage analysis demonstrated that exactly 131 cells die during normal *C. elegans* development. In the 1980s, researchers found that a mutation inactivating a single gene allows all 131 cells to live. Further research revealed that this gene is part of a pathway that controls and carries out apoptosis in a wide range of animals, including humans. In 2002, Brenner, Horvitz, and Sulston shared a Nobel Prize for their use of the *C. elegans* fate map in studies of programmed cell death and organogenesis.

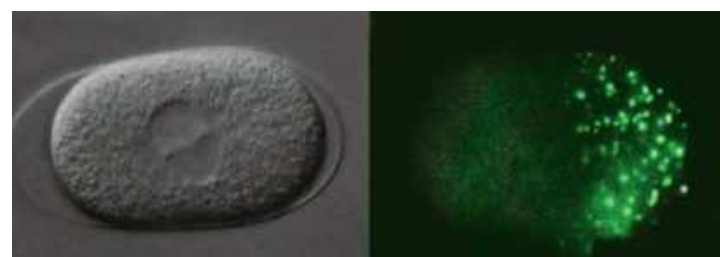
Having established fate maps for early development, scientists were positioned to answer questions about underlying



▼ **Figure 47.21 Partitioning of P granules during *C. elegans* development.** The differential interference contrast micrographs (left) highlight the boundaries of nuclei and cells through the first two cell divisions. The fluorescence micrographs (right) show identically staged embryos labeled with a fluorescent antibody specific for a P granule protein.



1 Newly fertilized egg



2 Zygote prior to first division



3 Two-cell embryo



4 Four-cell embryo

mechanisms, such as how the basic axes of the embryo are established, a process known as axis formation.

## Axis Formation

A body plan with bilateral symmetry is found across a range of animals, including nematodes, echinoderms, and vertebrates (see Concept 32.4). This body plan exhibits asymmetry along the dorsal-ventral and anterior-posterior axes, as